Problem 3

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Problem 1 *Evaluate the following sums:*

(a) $\sum_{k=1}^{4} k^5$

Explanation. $\sum_{k=1}^{4} k^5 = 1^5 + 2^5 + 3^5 + 4^5 = 1 + 32 + 243 + 1024 = 1300.$

(b) $\sum_{k=1}^{400} (5(k+1)^2 + 3)$

Explanation.

$$\begin{aligned}
\sum_{k=1}^{400} (5(k+1)^2 + 3) &= \sum_{k=1}^{400} (5(k^2 + 2k + 1) + 3) \\
&= \sum_{k=1}^{400} (5k^2 + 10k + 8) \\
&= \sum_{k=1}^{400} 5k^2 + \sum_{k=1}^{400} 10k + \sum_{k=1}^{400} 8 \\
&= 5\sum_{k=1}^{400} k^2 + 10\sum_{k=1}^{400} k + 8(400) \\
&= 5\left(\frac{400(400+1)(2(400)+1)}{6}\right) + 10\left(\frac{400(400+1)}{2}\right) + 3,200 \\
&= 5(200)(401)(267) + 10(200)(401) + 3,200 \\
&= 107,067,000 + 802,000 + 3,200 = 107,872,200
\end{aligned}$$